

Distance Formula & Simplifying Radicals

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Today's class topics — homework set. Show your work clearly.

Today we learned. The distance formula is nothing but the Pythagorean theorem in disguise. Visualize a right triangle between the two points — the horizontal leg is the x -difference, the vertical leg is the y -difference, and the hypotenuse is the distance.

Goal of this homework. Build the visual habit (Part A) and tighten radical simplification (Part B) — these two skills meet in every distance answer.

⚠ Watch your habits (Kenny's note from class). (1) *Simplify* $\neq$ *Solve* — Part B asks you to **simplify**, not to set things equal. (2) Don't invent steps that aren't there. Write what the problem actually says, then reduce.

Part A — Distance Formula

Find the distance between each pair of points. Leave answers in simplest radical form when needed.

1. $(1, 2)$ and $(6, 3)$

Distance = _____

2. $(0, -6)$ and $(7, 1)$

Distance = _____

3. $(-3, 4)$ and $(5, -2)$

Distance = _____

4. $(-5, 7)$ and $(-7, 8)$

Distance = _____

5. $(0, 0)$ and $(3, 4)$ (*classic — sketch the triangle!*)

Distance = _____

6. $(2, -1)$ and $(2, 6)$

Distance = _____

7. $(-4, 3)$ and $(8, 3)$

Distance = _____

8. A triangle has vertices $(0, 0)$, $(3, 0)$, $(0, 4)$. Find its perimeter.

Perimeter = _____

9. The points $(2, 5)$ and $(x, 5)$ are 6 units apart. Find both possible values of x .

$x =$ _____

10. The points $A(-1, 2)$, $B(4, 2)$, $C(4, 6)$ form a right triangle. Find the length of AC (the hypotenuse).

$AC =$ _____

Part B — Simplifying Radicals

Simplify completely. Pull out perfect squares; leave only what cannot be reduced inside the radical.

Quick check. For variables under the radical: even exponent = pulls out fully ($\sqrt{x^6} = x^3$); odd exponent = pull out the even part, leave one x inside ($\sqrt{x^9} = x^4\sqrt{x}$).

1. $\sqrt{196x^6}$

= _____

2. $\sqrt{24x^9}$

= _____

3. $\sqrt{180x^{36}y^{25}}$

= _____

4. $2x \sqrt{12x^{35}y^{16}}$

= _____

5. $\sqrt{50}$

= _____

6. $\sqrt{72x^4}$

= _____

7. $\sqrt{8x^3} \cdot 3\sqrt{12x}$

= _____

8. $\sqrt{45x^5y^3}$

= _____

9. $\sqrt{98x^7y^{10}}$

= _____

10. $\sqrt{8x^3} \left(3\sqrt{12x} - 4x\sqrt{18x^9} \right)$ (distribute, then simplify each piece)

= _____